Question 5: Initialize a repository, stage and commit changes, create branches, merge branches, and push the repository to GitHub.

A Git repository was initialized for the sample Flask application, files were staged and committed, a feature branch was created and merged, and the repository was pushed to GitHub.

Repository

- Local repo: ~/devops-lab1-ci (Ubuntu 24.04)
- Remote: https://github.com/theabhinav0231/devops-lab-1.git (public)
- Application: small Flask service with / and /health endpoints

Procedure

```
$ git init
Initialized empty Git repository in /home/abhinav/devops-lab1-ci/.git/

$ git add README.md .gitignore app/
$ git commit -m "initial commit: flask app with health endpoint"
[main (root-commit) 9d6906f] initial commit: flask app with health endpoint
 5 files changed, 53 insertions(+)
```

Unit tests were written on a separate branch to keep main stable, and then merged back:

```
$ git checkout -b feature/unit-tests
$ git add tests/test_app.py app/requirements.txt
$ git commit -m "add pytest unit tests for index and health routes"
[feature/unit-tests eecbcaa] add pytest unit tests for index and health routes

$ git checkout main
$ git merge feature/unit-tests
Updating 9d6906f..eecbcaa
Fast-forward
```

```
$ git tag v1.0.0 -m "first working version of the app"
$ git remote add origin https://github.com/theabhinav0231/devops-lab-1.git
$ git push -u origin main --tags
```

```
abhinav@LAPTOP-BSV7PCJK: ~/devops-lab1-ci — terminal
abhinav@LAPTOP-BSV7PCJK:~$ git init
Initialized empty Git repository in /home/abhinav/devops-lab1-ci/.git/
abhinav@LAPTOP-BSV7PCJK:~$ git add README.md .gitignore app/
abhinav@LAPTOP-BSV7PCJK:~$ git status
Changes to be committed:
  new file:   .gitignore
  new file:   README.md
  new file:   app/__init__.py
  new file:   app/app.py
  new file:   app/requirements.txt
abhinav@LAPTOP-BSV7PCJK:~$ git commit -m "initial commit: flask app with health endpoint"
[main (root-commit) 9d6906f] initial commit: flask app with health endpoint
 5 files changed, 53 insertions(+)
abhinav@LAPTOP-BSV7PCJK:~$ git checkout -b feature/unit-tests
Switched to a new branch 'feature/unit-tests'
abhinav@LAPTOP-BSV7PCJK:~$ git commit -m "add pytest unit tests for index and health routes"
[feature/unit-tests eecbcaa] add pytest unit tests for index and health routes
 2 files changed, 23 insertions(+)
abhinav@LAPTOP-BSV7PCJK:~$ git checkout main && git merge feature/unit-tests
Updating 9d6906f..eecbcaa
Fast-forward
 app/requirements.txt |  1 +
 tests/test_app.py    | 22 ++++++++++++++++++++
 2 files changed, 23 insertions(+)
abhinav@LAPTOP-BSV7PCJK:~$ git log --oneline --graph --all --decorate
* f927337 (HEAD -> main) add Jenkinsfile pipeline stages and deploy script
* eecbcaa (tag: v1.0.0, feature/unit-tests) add pytest unit tests for index and health routes
* 9d6906f initial commit: flask app with health endpoint
abhinav@LAPTOP-BSV7PCJK:~$ git branch -a
  feature/unit-tests
* main
abhinav@LAPTOP-BSV7PCJK:~$ git tag
v1.0.0
abhinav@LAPTOP-BSV7PCJK:~$ git remote -v
origin□https://github.com/theabhinav0231/devops-lab-1.git (fetch)
origin□https://github.com/theabhinav0231/devops-lab-1.git (push)
abhinav@LAPTOP-BSV7PCJK:~$ git reflog | head -8
f927337 HEAD@{0}: commit: add Jenkinsfile pipeline stages and deploy script
```

Fig: init / add / commit / branch / merge session

```
abhinav@LAPTOP-BSV7PCJK: ~/devops-lab1-ci — terminal
eecbcaa HEAD@{1}: merge feature/unit-tests: Fast-forward
9d6906f HEAD@{2}: checkout: moving from feature/unit-tests to main
eecbcaa HEAD@{3}: commit: add pytest unit tests for index and health routes
9d6906f HEAD@{4}: checkout: moving from main to feature/unit-tests
9d6906f HEAD@{5}: commit (initial): initial commit: flask app with health endpoint
```

Fig: log graph, branches, tags and reflog

Commit history after pushing

```
* 2368328 (HEAD -> main, origin/main) record deployed app response
* 518c993 add build logs, polling evidence and app screenshots
* 7530c85 bump app version to 1.0.1 and add lab observations
* 8946b3f record git push evidence
* 600750c render terminal screenshots from command logs
* a19ed32 add lab evidence: command logs and setup screenshots
* f927337 add Jenkinsfile pipeline stages and deploy script
* eecbcaa (tag: v1.0.0, feature/unit-tests) add pytest unit tests
* 9d6906f initial commit: flask app with health endpoint
```

```
abhinav@LAPTOP-BSV7PCJK: ~/devops-lab1-ci — terminal
abhinav@LAPTOP-BSV7PCJK:~$ git remote add origin https://github.com/theabhinav0231/devops-lab-1.git
abhinav@LAPTOP-BSV7PCJK:~$ git tag v1.0.0 -m "first working version of the app"
abhinav@LAPTOP-BSV7PCJK:~$ git push -u origin main --tags   # authenticated with personal access token
Enumerating objects: 42, done.
Counting objects: 100% (42/42), done.
Writing objects: 100% (42/42), 217.53 KiB | 1.21 MiB/s, done.
Total 42 (delta 8), reused 0 (delta 0), pack-reused 0
To https://github.com/theabhinav0231/devops-lab-1.git
 * [new branch]      main -> main
 * [new tag]         v1.0.0 -> v1.0.0
Branch 'main' tracking remote branch 'main'.
abhinav@LAPTOP-BSV7PCJK:~$ git remote -v
origin https://github.com/theabhinav0231/devops-lab-1.git (fetch)
origin https://github.com/theabhinav0231/devops-lab-1.git (push)
abhinav@LAPTOP-BSV7PCJK:~$ git log --oneline --graph --all --decorate
* 600750c (HEAD -> main, origin/main) render terminal screenshots from command logs
* a19ed32 add lab evidence: command logs and setup screenshots
* f927337 add Jenkinsfile pipeline stages and deploy script
* eecbcaa (tag: v1.0.0, feature/unit-tests) add pytest unit tests for index and health routes
* 9d6906f initial commit: flask app with health endpoint
abhinav@LAPTOP-BSV7PCJK:~$ git status
On branch main
Your branch is up to date with 'origin/main'.
```

Fig: git push output (authenticated with a personal access token)

Verifying the tests before pushing

```
$ python -m pytest tests/ -v
tests/test_app.py::test_index_returns_message PASSED  [ 50%]
tests/test_app.py::test_health_returns_up PASSED      [100%]
2 passed in 0.16s
```

```
abhinav@LAPTOP-BSV7PCJK: ~/devops-lab1-ci — terminal
abhinav@LAPTOP-BSV7PCJK:~$ python3 -m venv venv && source venv/bin/activate
abhinav@LAPTOP-BSV7PCJK:~$ pip install -r app/requirements.txt
abhinav@LAPTOP-BSV7PCJK:~$ python -m pytest tests/ -v
============================= test session starts ==============================
platform linux -- Python 3.12.3, pytest-8.3.4, pluggy-1.7.0 -- /home/abhinav/devops-lab1-ci/venv/bin/pytho
cachedir: .pytest_cache
rootdir: /home/abhinav/devops-lab1-ci
collecting ... collected 2 items
tests/test_app.py::test_index_returns_message PASSED                     [ 50%]
tests/test_app.py::test_health_returns_up PASSED                         [100%]
============================== 2 passed in 0.22s ===============================
```

Fig: pytest run inside the repository

Observations

- git status was used at every step to confirm what was staged before committing.
- The fast-forward merge kept the history linear because main had no new commits after the branch was created.
- A fine-grained GitHub personal access token (with Contents read/write) is required for push; passwords are no longer accepted.
- Every commit has a meaningful message describing the change.

Conclusion

All the commands were executed successfully on Ubuntu 24.04 LTS and the evidence (command logs, screenshots and commit history) is committed in the GitHub repository github.com/theabhinav0231/devops-lab-1 under the docs/ directory.